Quantitative analysis of Raman data of pesticide from real sample based on spiking neural networks

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ABSTRACT

In this study, we proposed a Spiking Neural Network (SNN) based framework for analyzing Raman spectral data. The model was trained to predict the concentration of three substances(Malachite Green, 4-MBA and Thiram) by analyzing their Surface-Enhanced Raman Scattering(SERS) spectra at various concentrations, an average prediction accuracy exceeding 92% was achieved. As for testing Malachite Green from real samples, the value of R^2 was significantly higher than several other comparative models, that including Convolutional Neural Networks(CNN) and Multilayer Perceptrons(MLP). Furthermore, the method was successfully applied for quantitative detecting pesticide from real sample. This work establishes that SERS spectral analysis using SNN models represents a feasible, efficient, and precise method for quantitative analysis.

Keywords: Meaching learning; Spiking neural networks; SERS; Quantitative prediction, pesticide; real sample

1. Introduction

Since its discovery, surface-enhanced Raman spectroscopy (SERS) has been a powerful analytical tool, with the significant enhancement of the Raman signal primarily attributed to two factors: charge transfer between the analyte and the metallic substrate, and localized surface plasmon resonance (LSPR) in metallic nanomaterials upon light excitation[1-3].This capability enables SERS to push the detection limit to the single-molecule level while maintaining its ability for facile and rapid identification of substance concentrations or materials, leading to widespread applications in food security analysis and various other fields[4-6].Additionally, machine learning algorithms can be integrated with SERS spectroscopy to leverage their powerful computational capabilities for rapid identification of Raman features in each spectrum and the construction of predictive models for swift recognition and analysis.

In recent years, deep learning technology has advanced rapidly, driving remarkable breakthroughs in various pattern recognition tasks, including fundamental image recognition and segmentation, object detection, and speech and text processing[7-10].Meanwhile, the integration of deep learning with Raman spectroscopy has demonstrated significant potential in spectral identification. For example, in the domain of convolutional neural networks (CNNs), a dense convolutional neural network (DenseCNN) with over 40 layers was proposed to enhance feature propagation, enabling the identification of Raman spectra from 32 liquid chemicals[10]. Additionally, a model combining residual networks with one-dimensional CNNs (1D-CNN) was developed for the analysis of common bacteria[11].Furthermore, a hybrid network integrating Savitzky-Golay convolutional smoothing with CNN was introduced to suppress noise and extract features in SERS spectra[12];In the field of long short-term memory (LSTM) networks, Wu et al. achieved qualitative and quantitative analysis of phthalates in extra virgin olive oil (EVOO) by integrating SERS with an LSTM neural network[13].Lin et al. rapidly identified dibenz(a,h)anthracene (DbA) and benzo(b)fluoranthene (BbF) in mussels using a similar integration of SERS and LSTM networks[14].

However, most current algorithms rely on empirical parameter adjustments — such as smoothing window width and baseline correction thresholds — during preprocessing, which can lead to the loss of valuable information15, 16. Moreover, continuous activation functions such as Softmax often struggle to effectively address the non-stationary characteristics of SERS signals[17-19].

In order to address these challenges, we propose a Spiking Neural Network (SNN) based framework. The third-generation spiking neural network (SNN) has advanced brain-inspired networks by drawing on neurophysiological insights, controlling neuronal activity through high/low membrane potentials, and discarding the complex computations of traditional deep neural networks. This enables ultra-low-power operation and event-driven processing feasible for neuromorphic hardware. A spiking neural network consists of spiking neurons, each of which fires an output spike only when its membrane potential is charged above a certain threshold [20-23]. The generated spike is propagated into the neurons in the next layer and increases/decreases their membrane potentials. In this manner, the communication between neurons is performed by spikes.

The rate encoding was employed in SNN model[24, 25], where Poisson-distributed random numbers are used to sample pulses over T time intervals for encoding, enabling direct conversion of raw spectral data into spatio-temporal pulse sequences. Thus avoiding systematic errors introduced by traditional preprocessing methods. The bio-inspired Leaky Integrate-and-Fire (LIF) neuron model is used to construct the output layer[26-28], and the Spike-Timing-Dependent Plasticity (STDP) mechanism captures the temporal evolution of spectral features. To address non-differentiability of the pulse signal, an error backpropagation mechanism based on the Surrogate Gradient method is employed to optimize network parameters, enabling parameter optimization while maintaining biological plausibility[28, 29]. The model was ultimately tested on three types of substances for concentration analysis and selected pesticides for real sample testing, demonstrating its efficiency and applicability.

2. Materials and methods

2.1. Instruments and Reagents

The Raman spectra were collected using a portable Raman spectrometer (BWS465-785S, B&W-Tek, USA) with an excitation wavelength of 785 nm and a Raman shift range of 110-2000cm^{-1}.The UV-Vis-NIR absorption spectra were recorded using an Agilent Cary series UV-Vis-NIR spectrophotometer (Agilent Technologies, USA).An SU8010 scanning electron microscope (SEM, Hitachi, Japan), a DF-101S heating-type magnetic stirrer (Gongyi Yuhua Instrument Co., Ltd.), and a computer with an i7-11390H processor (3.40 GHz) and 16.0 GB RAM were also utilized in the experiments.

Thiram ($C_6H_{12}N_2S_4$, 97%) and silver nitrate ($AgNO_3$, 99%) were purchased from Innochem Technology Co., Ltd.Sodium citrate ($C_6H_5Na_3O_7$, 99%) was obtained from Sinopharm Chemical Reagent Co., Ltd.4-MBA($C_7H_6O_2S$, 90%) and MG ($C_{23}H_{25}ClN_2$, AR) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd.

2.2. Preparation of silver nanoparticles

Glassware used for silver nanoparticle (AgNPs) synthesis was pre-cleaned with aqua regia (HNO_3:HCl, 1:3 v/v) to remove metal contaminants other impurities. AgNPs were synthesized sodium citrate reduction following literature procedure[30].Briefly, 0.0342 g of $AgNO_3$ was dissolved in 200 mL of ultrapure water, followed by heating under constant stirring at 250 r/min for 25 min until refluxing. Then, 4 mL of 0.01 mol/L sodium citrate solution was rapidly added, and the mixture was maintained at 100°C for an additional 20 min. After cooling to room temperature, the colloidal solution was centrifuged at 1500 r/min for 5 min, and the lower turbid layer was carefully collected using a pipette.

2.3. Experimental Model

2.3.1. Brief introduction to Backpropagation network

In regression tasks performed by the model, the core process relies on an interconnected multi-layer neural network architecture. The backpropagation (BP) network, a gradient-based optimization method, plays a central role by adjusting model parameters through backpropagation to minimize the error function between predicted and target outputs during forward propagation. This adjustment mechanism

enables the model to iteratively refine its predictive capabilities, gradually enhancing the output of the error function is systematically minimized through successive training cycles. By continuously aligning the results with the values via backpropagation, the model progressively adapts its internal parameters to better capture the patterns data patterns, achieving increasingly precise prediction predictions. dynamic optimization process ensures strengthens model's generalization ability ability, it to deliver more reliable and accurate forecasts across diverse input scenarios.

2.3.2. Spiking neural network

SNN is a type of artificial neural network that emulate the brain's biological decision-making processes through spiking neurons. In the brain, neurons communicate by propagating electrical signals. They accumulate membrane potentials by integrating incoming spike potentials. When the membrane potential exceeds a certain threshold, the neuron generates a new spike potential that is transmitted to the next neuron.Fig.1 illustrates the process of classifying Surface-Enhanced Raman Scattering (SERS) spectra using an SNN. First, spike trains are generated through spike encoding based on the provided SERS spectra, resulting in a spike matrix. This spike matrix serves as the input to the SNN. Neurons within the SNN compute their membrane potentials using the input spike matrix. When a neuron's membrane potential surpasses a predefined threshold, it emits a new spike signal. Through this propagation mechanism, the model classifies the SERS spectra by analyzing the firing frequency of the spikes.

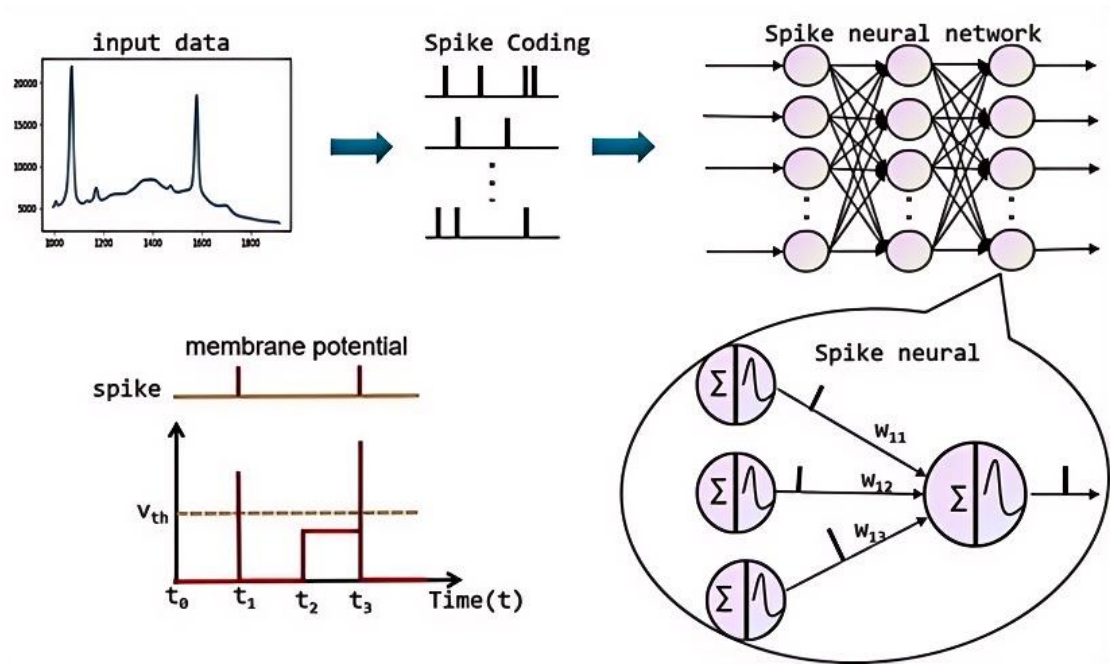

Fig.1 Flowchart of SNN Model

2.3.3. Rate coding

A good encoding system is a prerequisite for implementing a spiking neural network (SNN). One of the most commonly used encoding schemes in SNNs is rate coding, which represents signals by leveraging the frequency of spike discharges. In the rate coding scheme, the spike firing rate F_i of the i-th neuron can be formulated as follows:

$$F_i = \frac{\sum_{t=1}^{T} S_i(T)}{T}.$$

0-1 normalization is a data processing method that aims to restrict all data within the range of 0 to 1 to reduce calculation errors. It can be expressed as follows:

$$X_{new} = \frac{X - X_{(min)}}{X_{(max)} - X_{(min)}}.$$

As shown in the Fig.2,a Poisson distribution is used for pulse encoding of SERS spectra. For each SERS spectrum, a random number R is independently generated at each time step t, and pulse encoding is performed by comparing the SERS peak intensity at the current position with the random number.The pulse encoding formula for the i-th neuron at time step t can be as follows:

$$S_i(t) = \begin{cases} 1, & \text{IF } R \le X \\ 0, & \text{otherwise} \end{cases}.$$

Where X represents the SERS spectral data after 0-1 normalization.

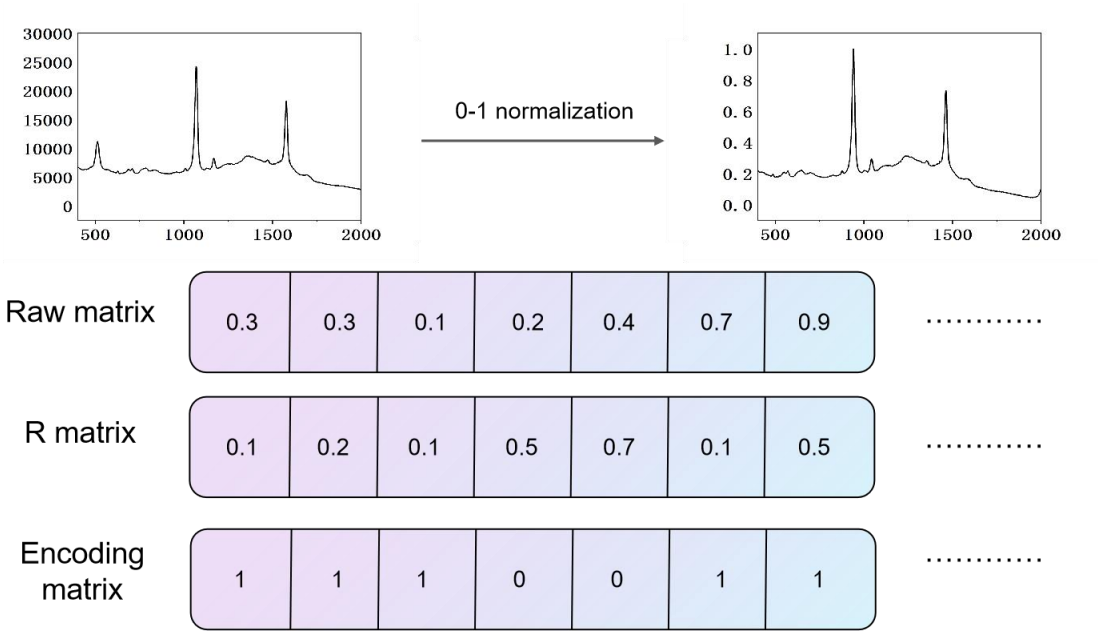

Fig.2 Spike Coding

2.3.4. The LIF Model

Due to the efficient temporal capture and information processing capabilities of SNN, we adopt the LIF neuron model to construct the SNN.The following formula effectively represents the membrane potential update equation of the neuron:

$$V[n] = V[n-1] + \frac{\Delta t}{\tau}(I[n] - (V[n-1] - V_{reset}))$$

V[n] is the membrane potential function at time n. V_{reset} represents the resting membrane potential of the neuron. τ is a constant that characterizes the charging and discharging rates of the neuron's membrane potential. I[n] is the pulse input function, which corresponding to the data content of our SERS spectra.

When the membrane potential V[n] exceeds the threshold potential V_{th}, the neuron is activated and triggers a spike.V_{reset} is the reset potential to which the membrane potential returns after the neuron fires.

$$S[n] = \begin{cases} 1, & \text{IF } V[n] \ge V_{th} \\ V_{reset}, & \text{otherwise} \end{cases}$$

Each neuron simultaneously receives multiple pulse signals. These signals occur

independently over t time steps and collectively induce a sustained modification of the membrane potential through the membrane potential update function.

2.3.5. Surrogate Gradient of SNN

For artificial neural networks (ANNs), backpropagation is a fundamental process for improving prediction accuracy. By applying the chain rule, it adjusts weights in a backward pass from the output layer to the input layer, enabling efficient optimization of millions of parameters. In contrast, training SNN presents challenges due to their discrete and non-differentiable nature. Consequently, surrogate gradient methods have become one of the most common approaches for parameter optimization in SNN.

For training the SNN, with the loss function denoted as L, the update process of SNN parameters can be expressed according using chain rule as follows:

$$\frac{\partial L(\omega)}{\partial \omega} = \sum_{n} \frac{\partial L(\omega)}{\partial S[n]} \frac{\partial S[n]}{\partial V[n]} \frac{\partial V[n]}{\partial I[n]} \frac{\partial I[n]}{\partial \omega}$$

where $\frac{\partial S[n]}{\partial V[n]}$ is the gradient of the spiking function. Due to I set Vreset to 0, the gradient of the spiking function is nonzero only when $V[n]>V_{th}$, which is unsuitable for gradient descent-based parameter learning. To address this issue, surrogate gradient methods replace the original discontinuous gradient with a smooth function. A typical example is the Arctan function (also known as the ATan surrogate function), which provides a differentiable approximation for training spiking neural networks.

$$\mathrm{ATan} = \frac{1}{\pi}\arctan(\frac{\pi}{2}\mathrm{ax}) + \frac{1}{2}$$

$$\frac{\partial S[n]}{\partial V[n]} = \frac{a}{2(1+(\frac{\pi}{2}\mathrm{ax})^2)}$$

The ATan surrogate gradient is computed based on the distance between the membrane potential V[t] and the threshold V_{th}, where α is a hyperparameter that determines the slope of the function. Then, we can update the gradient like a traditional BP:3.Results and discussion

$$\omega = \omega - \eta \frac{\partial L(\omega)}{\partial \omega}$$

3. Results and discussion

3.1. Characterization of Silver Nanoparticles

SEM images are primarily utilized to assess the size distribution of AgNPs. As illustrated in Fig.3(a) ,the predominant morphologies of the AgNPs are near-spherical and columnar. Although some particles exhibit slight surface roughness, their overall shapes remain clearly distinguishable. The measured diameters of the particles range from approximately 50 to 70 nm.

The ultraviolet (UV) spectrum plays a crucial role in confirming the synthesis of silver nanoparticles. As shown in Fig.3(b), a characteristic absorption peak appears at 410 nm, attributed to the surface plasmon resonance (SPR) phenomenon. The presence of this peak provides clear evidence that the synthesis of AgNPs was successful.

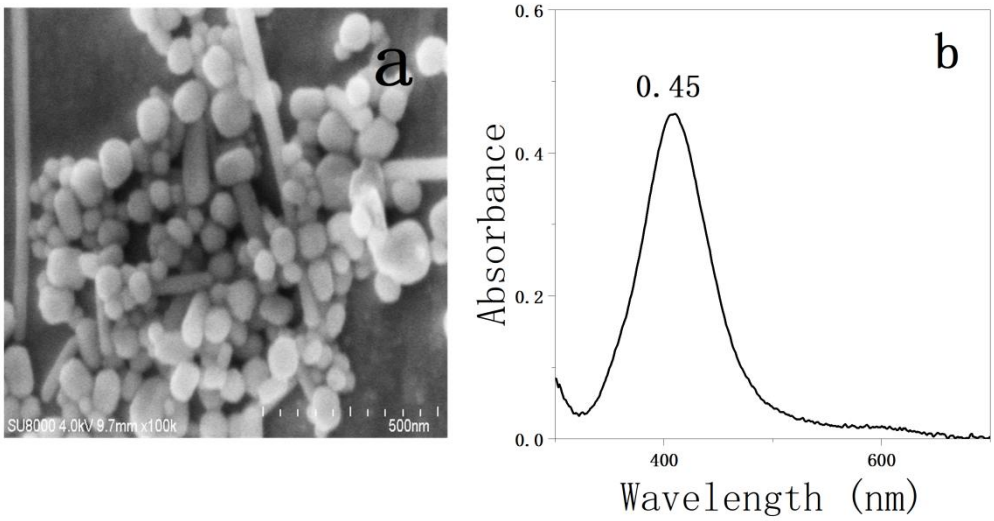

Fig.3 (a) SEM image of AgNPs, (b) UV-Vis absorption spectrum of AgNPs.

3.2. Experiments of the Algorithm

Quantitative analysis has garnered significant research interest in the field of SERS. Due to the nonlinear relationship between SERS spectra and substance concentration, most studies modify the original SERS data to satisfy the model's assumptions, which often results in the loss of some characteristic information. In this context, SNN are employed as prediction models for SERS. This approach utilizes pulse encoding to capture the characteristic information of SERS spectra and leverages LIF neurons for concentration prediction and classification.

3.3. Acquisition of SERS Spectra

First, solutions of MG, 4-MBA, and thiram at various concentrations were prepared as training and testing reagents, with concentrations ranging from 1 ppm to 1000 ppm. Subsequently, 2 μ L of each solution was individually dropped onto a commercial silica gel plate, followed by the application of 2 μ L of AgNPs onto the plate. Then, 10 sets of SERS spectra were collected from different sampling points for each sample at varying concentrations.Fig.4 presents the relative standard deviations (RSDs) of the three sets of collected SERS intensities.Fig.5(a) shows that the SERS spectrum of 4-MBA.The peak at 1084 cm^{-1} arises from the coupling of the benzene ring skeletal vibration and the C–S single-bond stretching vibration, while the peak at 1580 cm^{-1} is attributed to the uncoupled normal vibration of the benzene ring under symmetric conditions [31].Fig.5(b) shows that the SERS characteristic peak of thiram at 556 cm^{-1} originates from the intramolecular C–N stretching and CH_3 rocking vibrations, while the peak at 1310 cm^{-1} is attributed to C–N stretching and symmetric deformation of CH_3 groups in thiram molecules[32].As shown in Fig.5(c),the SERS characteristic peak of MG at 798 cm^{-1} corresponds to the out-of-plane vibration of the C–H group; the peak at 916 cm^{-1} arises from the ring skeletal vibration with radical orientation; the peak at 1173 cm^{-1} is attribute to the in-plane vibration of the aromatic ring; the peak at 1396 cm^{-1} results from the symmetric stretching vibration of the C–N group; and the peak at 1618 cm^{-1} is assigned to the stretching vibration of the C=C ring [33].

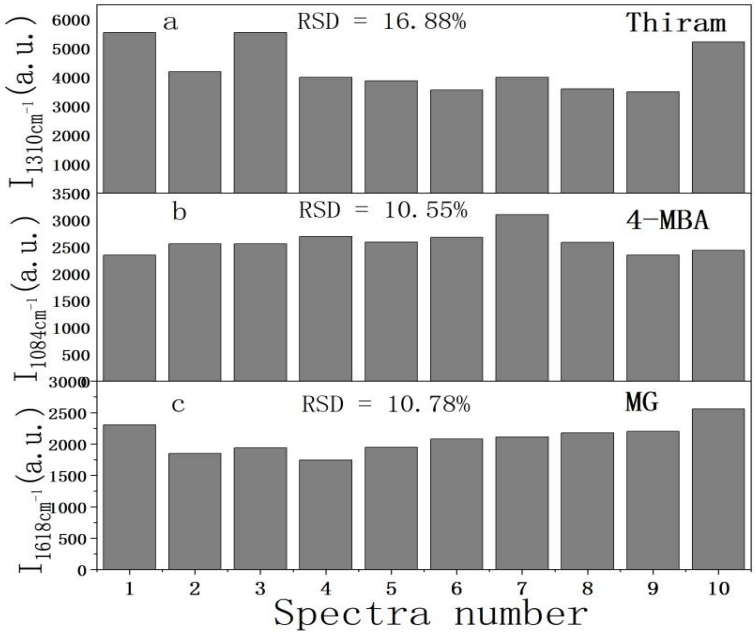

Fig.4 the different RSDs(a,b,c) of thiram,4-MBA, and MG.

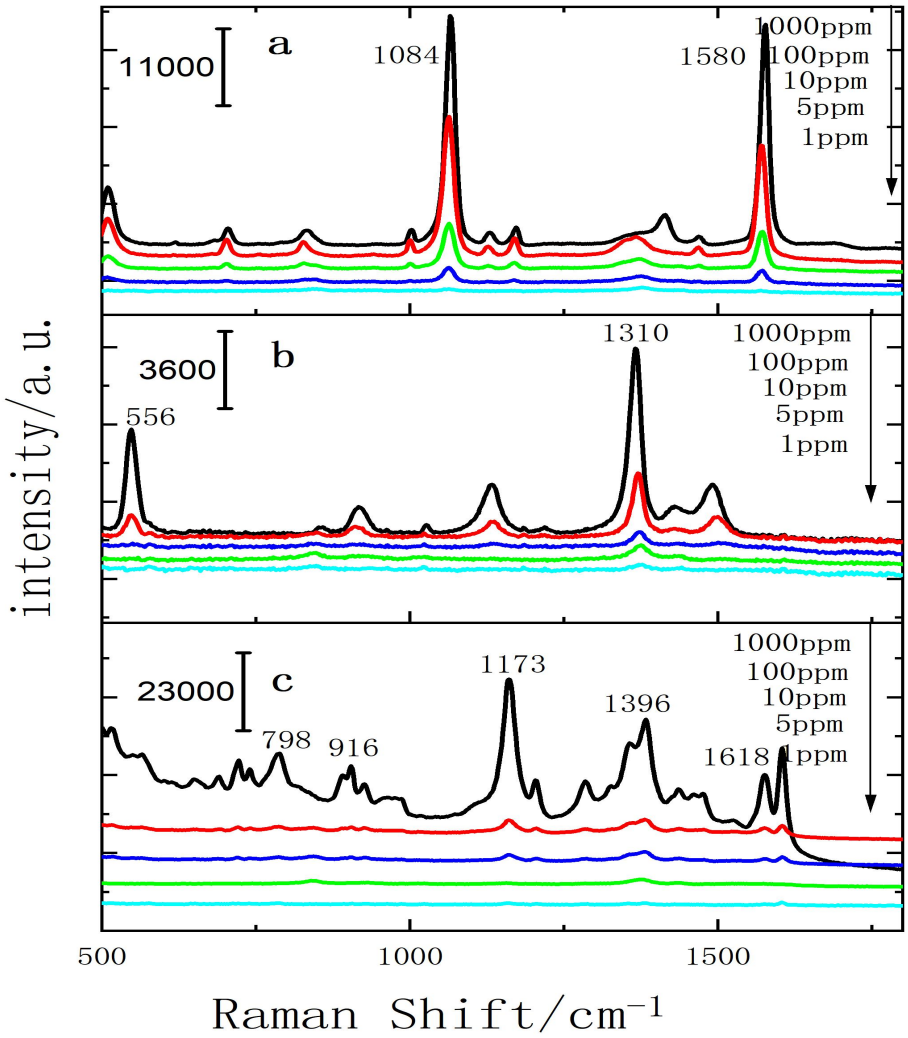

Fig.5 The SERS spectra (a, b, c) of 4-MBA, thiram, and MG at different

concentrations

3.4. Training Modle

Then,the SERS spectral data were processed using 0-1 normalization and spike encoding. After generating the corresponding spike encoding matrix, 104 pieces of encoded information were randomly selected as the training set, and 29 pieces as the test set. With the SNN parameters set as T=100 time steps, epoach=100, and τ=5.1, the 104 training samples were fed into the network for training.At the same time, we constructed CNN and MLP models and selected 104 data samples for training.The architectures of the three models are shown in Table.1.

Fig.5 illustrates the models' performance during training on the test data.The results showed that the predicted concentrations were in good agreement with the true values, demonstrating relatively high prediction accuracy. For the SERS spectra of the test set, the concentrations predicted by the SNN model were essentially identical to the actual values. To evaluate the model's performance, the coefficient of determination (R-squared, R^2), mean squared error (MSE), and mean absolute error (MAE) were used to assess the prediction accuracy of the models.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(\bar{y} - y_i)^2} \quad MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \quad MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right|$$

where y_i is the true value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the true values, and n is the number of samples.

As shown in the Table.2, we compared the SNN model with CNN and MLP models.The SNN model outperformed both CNN and MLP models, achieving higher R^2 values and lower MSE and MAE metrics. This superior predictive ability stems from its inherent spike-based encoding, which captures richer spectral features from SERS data and more comprehensively leverages the nonlinear characteristics of SERS spectra. The implementation of the SNN model enables faster and more convenient quantitative analysis in SERS applications

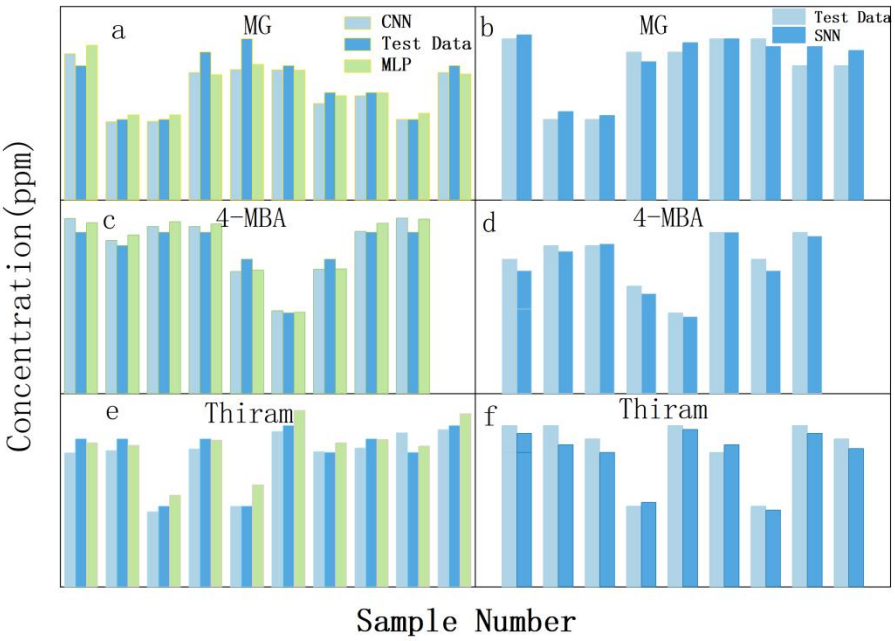

Fig.6 The histograms compare the real concentrations with the predicted concentrations from the CNN testing set and MLP testing set (a, c, e), as well as from the SNN testing set (b, d, f)

Table.1 Structures of the SNN,CNN and MLP models

Model name	Model components
CNN	Conv1d(16,2) Flatten Dense(800) Dense(1)
MLP	Dense(800) Dense(1)
SNN	Linear(800,1) LIF

Table.2 Comparison of Results among SNN, CNN, and MLP Models.

Model name	Detectable Substance	R-squared	Mean Squared Error	Mean Absolute Error
CNN	4-MBA	0.8704	0.0017	0.0331
	thiram	0.8726	0.0003	0.0153
	MG	0.8781	0.0015	0.0331
MLP	4-MBA	0.8471	0.0012	0.0282
	thiram	0.8672	0.0004	0.0172
	MG	0.6285	0.0025	0.0480
SNN	4-MBA	0.9492	0.0005	0.0222
	thiram	0.8855	0.0008	0.0267
	MG	0.9037	0.0002	0.0133

3.5. Application of SNN in Real Samples

The SNN model was employed for the quantitative analysis of SERS spectra of MG and thiram in real samples. As broad-spectrum fungicides, MG and Thiram have been widely applied in marine aquaculture and agricultural production respectively, serving to inhibit the excessive proliferation of harmful pathogens and microorganisms. However, the extensive use of MG and Thiram often leads to their excessive residues in fish and agricultural products respectively, posing potential risks to food safety and public health. It is essential to quantitatively detect their concentrations in real samples. The SERS technique was employed to detect the concentration of MG in real lake water samples and the concentration of Thiram on the surface of real apple peels. Additionally, an accurate quantitative model, the SNN, was developed to analyze the SERS spectra and evaluate the levels of MG and Thiram in the respective samples. Lake water samples were collected from a local lake, and apple samples were purchased from local supermarkets. Then, 0.0046 g of MG was dissolved in the lake water to prepare a stock solution with a concentration of 0.01 mol/L. This stock solution was then serially diluted with lake water to obtain MG solutions at concentrations of 100 ppm, 10 ppm, and 1 ppm. For each concentration, 2 μL of the solution was deposited onto a commercial silica gel plate, followed by the addition of 2 μL of AgNPs for SERS spectral acquisition, as illustrated in the

Fig.7(a).Then, 0.026 g of thiram was dissolved in ethanol to prepare a stock solution with a concentration of 0.01 mol/L. This stock solution was then diluted to obtain thiram solutions at concentrations of 1000 ppm, 100 ppm, and 10 ppm. For each concentration, 2 μL of the solution was sprayed onto the surface of cleaned apples, followed by the addition of 2 μL of AgNPs for SERS spectral acquisition, as illustrated in the Fig.7(c).The SERS spectral data were processed using SNN, CNN, and MLP models, respectively. The predicted concentrations of MG and thiram obtained from these models in real samples are shown in Fig.7(b,d). It can be observed that the fitting curve of the SNN model is closest to the original curve. Specifically, its R^2 value for MG reaches 0.96, outperforming those of the CNN and MLP models (0.92 and 0.86, respectively). For thiram, the R^2 value of the SNN model is 0.95, which is higher than the 0.91 of the CNN model and 0.83 of the MLP model.

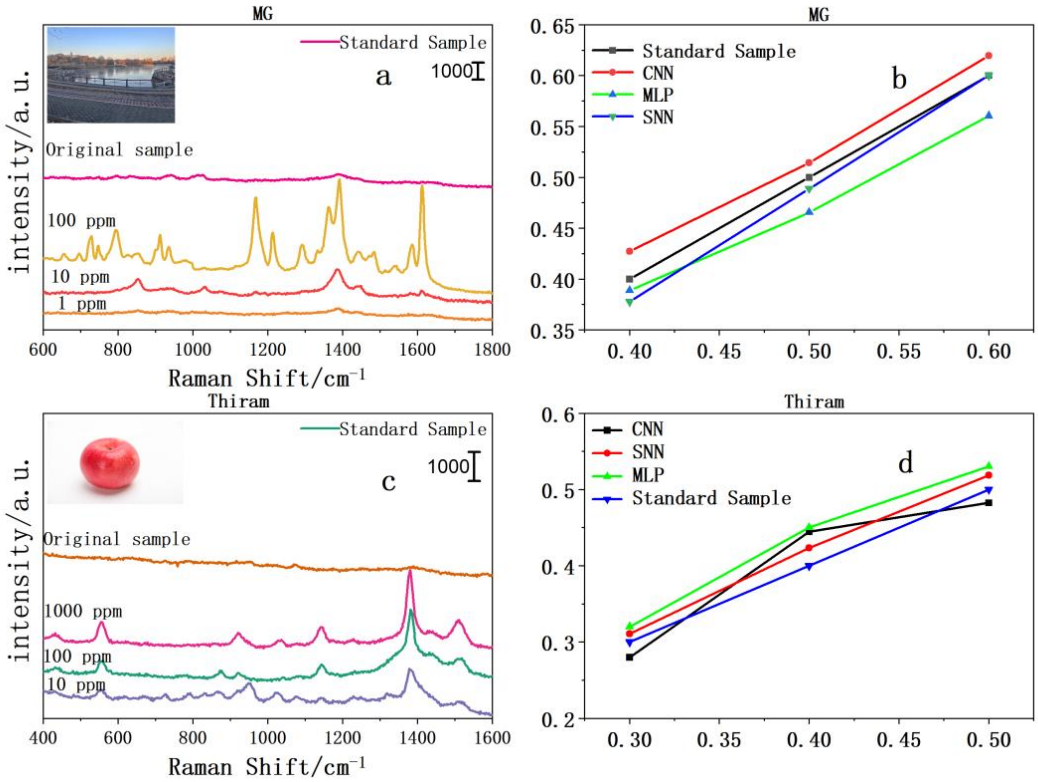

Fig.7 The SERS spectra(a,c) of MG and thiram with different concentrations and the prediction accuracy graph of SNN, CNN, and MLP models (b,d).

3.6. Conclusion

In this study, we developed a novel neurodynamics-based artificial intelligence

model employing a SNN for quantitative SERS spectral detection. The model incorporates a spike encoding layer with LIF neurons to rapidly and accurately analyze spectral features, enabling high-precision predictive capabilities. These mehod was validated by using pesticides in real sample, the model demonstrates robust extraction of authentic SERS features and a strong correlation between concentration and prediction. The experimental results indicate that the SNN outperforms CNN and MLP, achieving higher R^2 values and lower MAE and MSE, reflecting predictions closer to true values.These findings position the SNN as a more precise, convenient, and innovative approach for SERS spectral analysis and quantitative detection.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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